



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

DIGITAL FORENSICS

A PIXEL 3 IMAGE

OVERVIEW

- Pixel 3 Image Description
- File System Structure
- Understand the Apps/Package

PIXEL 3 IMAGE DESCRIPTION

How was the device image acquired?

ABOUT THE PIXEL 3 IMAGE

- Created by Joshua Hickman
- The Pixel device is first rooted
 - using the Team Win Recovery Project (TWRP)
 - TWRP enables users to install custom ROMs, root devices, and restore backups easily
- Not a real “dd image”
 - is a logic copy
- The image acquisition process



DOWNLOAD PIXEL 3 IMAGE

```
kali@kali: ~/pixel3 90x35

(kali㉿kali)-[~]
$ mkdir pixel3

(kali㉿kali)-[~]
$ cd pixel3

(kali㉿kali)-[~/pixel3]
$ wget -q https://digitalcorpora.s3.amazonaws.com/corpora/mobile/android_10/Non-Celebri
te%20Extraction/Pixel%203.zip

(kali㉿kali)-[~/pixel3]
$ ls Pixel\ 3.zip -l
-rw-r--r-- 1 kali kali 5247820897 Nov 21 22:14 'Pixel 3.zip'
```

A stock image of Android 10

VERIFY HASHES

```
kali@kali: ~/pixel3 90x35
(kali@kali)-[~/pixel3]
$ hashdeep Pixel\ 3.zip
%%% HASHDEEP-1.0
%%% size,md5,sha256,filename
## Invoked from: /home/kali/pixel3
## $ hashdeep Pixel 3.zip
##
5247820897,9cc37ebbbc4e918ee5427de1fe1deecc,ca6918ef8b20486b6a5ded15609ac51318f377829480f9
3be3ba15364a8aa00a,/home/kali/pixel3/Pixel 3.zip
```

SHA2-256: ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a

UNZIP TO GET PIXEL 3 IMAGE

```
kali@kali: ~/pixel3 90x35

(kali@kali)-[~/pixel3]
$ unzip -q Pixel\ 3.zip

(kali@kali)-[~/pixel3]
$ ls -l
total 5124836
drwxr-xr-x 11 kali kali      4096 Feb 15  2020 'Pixel 3'
-rw-r--r--  1 kali kali 5247820897 Nov 21 22:14 'Pixel 3.zip'
```



UNZIP TO GET PIXEL 3 IMAGE

```
kali@kali: ~/pixel3 90x35

(kali@kali)-[~/pixel3]
$ unzip -q Pixel\ 3.zip

(kali@kali)-[~/pixel3]
$ ls -l
total 5124836
drwxr-xr-x 11 kali kali      4096 Feb 15  2020 'Pixel 3'
-rw-r--r--  1 kali kali 5247820897 Nov 21 22:14 'Pixel 3.zip'
```



FILE SYSTEM STRUCTURE

Understand the locations may contain forensics data

list root directory

```
kali@kali: ~/pixel3 112x43
(kali@kali)-[~/pixel3]
$ ls -l Pixel\ 3
total 104
drwxr-xr-x 16 kali kali 4096 Feb 15 2020 apex
drwxr-xr-x 25 kali kali 4096 Feb 15 2020 data
-rw-r--r-- 1 kali kali 1653 Dec 31 2008 init.envIRON.rc
-rw-r--r-- 1 kali kali 33402 Dec 31 2008 init.rc
-rw-r--r-- 1 kali kali 7690 Dec 31 2008 init.usb.configfs.rc
-rw-r--r-- 1 kali kali 5649 Dec 31 2008 init.usb.rc
-rw-r--r-- 1 kali kali 563 Dec 31 2008 init.zygote32.rc
-rw-r--r-- 1 kali kali 981 Dec 31 2008 init.zygote64_32.rc
drwxr-xr-x 4 kali kali 4096 Feb 15 2020 metadata
drwxr-xr-x 3 kali kali 4096 Feb 15 2020 mnt
drwxr-xr-x 10 kali kali 4096 Feb 15 2020 product
drwxr-xr-x 3 kali kali 4096 Feb 15 2020 res
drwxr-xr-x 3 kali kali 4096 Feb 15 2020 sbin
drwxr-xr-x 12 kali kali 4096 Feb 15 2020 system
-rw-r--r-- 1 kali kali 2608 Dec 31 2008 ueventd.rc
drwxr-xr-x 16 kali kali 4096 Feb 15 2020 vendor
```

device

- To speed up security updates
 - Android Pony EXpress (APEX) is a container format introduced in Android 10 that is used in the install flow for lower-level system modules.
 - This format facilitates the updates of system components that don't fit into the standard Android application model.

run commands(rc): OS boot script

mount point for other file systems

Resource, e.g., layout, color

binaries for several important daemons.

Android OS (AOSP), exclude Google services

/SYSTEM: ANDROID OS PROJECT (AOSP)

```
kali@kali: ~/pixel3
(kali@kali)-[~/pixel3]
$ ls -l "Pixel 3/system"
total 56
drwxr-xr-x  2 kali kali  4096 Feb 15  2020 apex
drwxr-xr-x  4 kali kali  4096 Feb 15  2020 app
drwxr-xr-x  4 kali kali  4096 Feb 15  2020 bin
-rw-r--r--  1 kali kali  1970 Dec 31  2008 build.prop
drwxr-xr-x 14 kali kali  4096 Feb 15  2020 etc
drwxr-xr-x  2 kali kali 16384 Feb 15  2020 fonts
drwxr-xr-x  5 kali kali  4096 Feb 15  2020 framework
drwxr-xr-x  2 kali kali  4096 Feb 15  2020 lib
drwxr-xr-x  2 kali kali  4096 Feb 15  2020 lib64
drwxr-xr-x  6 kali kali  4096 Feb 15  2020 priv-app
drwxr-xr-x  7 kali kali  4096 Feb 15  2020 usr
```

"rooting" is required to modify the contents of this partition.

Android OS pre-installed bloatware apps, often can't uninstall

executable binaries, daemons or bash

App Framework

native libraries (C/C++; DLL)

- Privileged apps, apps that are eligible for system permissions

all system-wide, read-only files installed by (or provided by) the OS

It is less likely that you need to investigate the folder

/DATA

Contains subfolders
with **forensic data**

- all installed Android apps
 - native apps or
 - 3rd party
- all system config/log
- all app-generated data

```
kali@kali: ~/pixel3 102x39
(kali@kali)-[~/pixel3]
$ ls -l "Pixel 3/data"
total 108
drwxr-xr-x  3 kali kali 4096 Feb 15 2020 adb
drwxr-xr-x  4 kali kali 4096 Feb 15 2020 apex
drwxr-xr-x 83 kali kali 12288 Feb 15 2020 app
drwxr-xr-x  3 kali kali 4096 Feb 15 2020 backup
drwxr-xr-x  2 kali kali 4096 Feb 15 2020 cache
drwxr-xr-x  4 kali kali 4096 Feb 15 2020 dalvik-cache
drwxr-xr-x 103 kali kali 4096 Feb 15 2020 data
drwxr-xr-x  3 kali kali 4096 Feb 15 2020 drm
drwxr-xr-x  2 kali kali 4096 Feb 15 2020 magisk_backup
drwxr-xr-x  3 kali kali 4096 Feb 15 2020 media
drwxr-xr-x 20 kali kali 4096 Feb 15 2020 misc
drwxr-xr-x  3 kali kali 4096 Feb 15 2020 misc_ce
drwxr-xr-x  2 kali kali 4096 Feb 15 2020 nfc
drwxr-xr-x  2 kali kali 4096 Feb 15 2020 property
drwxr-xr-x  2 kali kali 12288 Feb 15 2020 resource-cache
drwxr-xr-x 14 kali kali 4096 Feb 15 2020 system
drwxr-xr-x  3 kali kali 4096 Feb 15 2020 system_ce
drwxr-xr-x  4 kali kali 4096 Feb 15 2020 system_de
drwxr-xr-x  2 kali kali 4096 Feb 15 2020 tombstones
drwxr-xr-x  3 kali kali 4096 Feb 15 2020 unencrypted
drwxr-xr-x  4 kali kali 4096 Feb 15 2020 user_de
drwxr-xr-x 12 kali kali 4096 Feb 15 2020 vendor
drwxr-xr-x  3 kali kali 4096 Feb 15 2020 vendor_ce
```

downloaded .apk

installed apps/generated data

system config/logs

DATA/SYSTEM SETTINGS (DON'T CONFUSE WITH /SYSTEM)

- Contains system **configuration** files critical to maintaining device's state.
- Contains system state **logs**
- Access to the directory is restricted by the system apps only, so if the device is not rooted the folder content can't be seen.
- for newer devices: **/data/system_xx/<userID>**

```
kali@kali: ~/pixel3 125x34
(kali@kali)-[~/pixel3]
$ ll 'Pixel 3/data/system'
appops.xml      dropbox/      netstats/     packages-warnings.xml  slice/
battery-history/ entropy.dat   notification_log.db    packages.xml          sound_model.db
batterystats.bin graphicsstats/ notification_policy.xml package-usage.list     st_sound_model.db
batterystats-checkin.bin install_sessions.xml overlays.xml          package-watchdog.xml  sync/
batterystats-daily.xml job/          package_cache/        recoverablekeystore.db uiderrors.txt
cachequota.xml  last-header.txt package-cstats.list   screen_on_time        usagestats/
device_policies.xml locksettings.db package-dcl.list      shortcut_service.xml  users/
diskstats_cache.json log-files.xml  package-dex-usage.list watchlist_report.db
display-manager-state.xml netpolicy.xml  packages.list         watchlist_settings.xml
```

- ***appops.xml*** - a configuration file used by the AppOps service which controls application permissions;
- ***called_pre_boots.dat*** - a file used by ActivityManager to hold pre-boot broadcast receivers;
- ***batterystats.bin*** - a binary used by the BatteryStats service to keeps power statistics of application installed;
- ***gesture.key*** - lockscreen pattern;
- ***inputmethod/*** - a directory used to store IME configuration.
Contains subtypes.xml with IME locales, keyboard layout sets etc;
- ***locksettings.db*** - a SQLite database containing the lock screen settings;
- ***netpolicy.xml*** - a configuration file used by the NetworkPolicyManagerService;
- ***netstats/*** - a directory used to hold NetworkStatsService statistics;

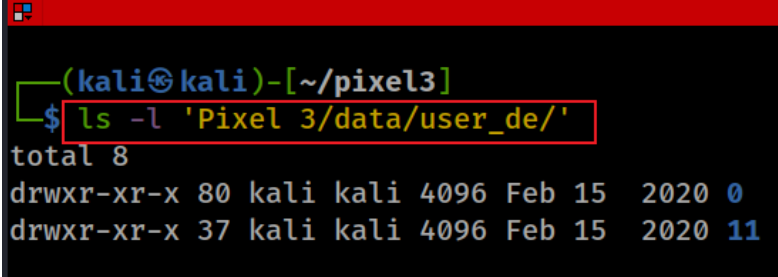
- ***packages.list*** - a file used by the PackageManager service containing the list of all packages (apks) installed in the system (same as executing `pm list packages` within shell);
- ***packages.xml*** - a file used by the PackageManager service to hold metadata of all installed packages, like UIDs, requested permissions etc;
- ***profiles.xml*** - a configuration file to keep metadata of device profiles including sound mode, airplane mode, vibration etc.;
- ***password.key*** - password hash;
- ***usagestats/*** - a directory used to store files for the UsageStats service. Contains `usage-history.xml` that holds info on all apps and its components with the last time a user used them;
- ***users/*** - a directory used for multi-user support. Contains `userlist.xml` which keeps info on all users in the system (0 - is a device owner/primary user); e.g. a sub-directory `users/0/` holds info on the user account (`accounts.db`), widgets (`appwidgets.xml`), wallpaper (`wallpaper_info.xml`) etc.

PRE-INSTALLED/DOWNLOADED APPS

- Pre-installed bloatware with ROM
 - **/system/app**: camera, settings, messages, Google Play Store, etc.
 - **/system/priv-app**: implicitly granted system privilege
 - **/custpack/app**
- downloaded **.apk** from Google Play Store
 - **/data/app**
- If moving apps to an SD card
 - Apps will be encrypted in **/mnt/sdcard/.android_secure**.
 - Android decrypts them at runtime and stores a decrypted copy on in **/mnt/asec/tmpfs**
 - Need to use a card reader and attach the SD card to a PC to see the files
 - **.asec** (Android Secure External Cache) instead of **.apk** -- from which you will get the connection to the name **/mnt/asec**

USER INSTALLED APPS

- not tight to specific users:
`/data/data/<package_name>`
- associate with specific users:
 - `/data/user/<userid>` (is a symlink to `/data/data`)
 - `/data/user_de/<userid>` for newer devices
- External storage for user app-generated data
 - `/mnt/sdcard/Android/data/<package_name>`



```
(kali㉿kali)-[~/pixel3]
$ ls -l 'Pixel 3/data/user_de/'
total 8
drwxr-xr-x 80 kali kali 4096 Feb 15 2020 0
drwxr-xr-x 37 kali kali 4096 Feb 15 2020 11
```

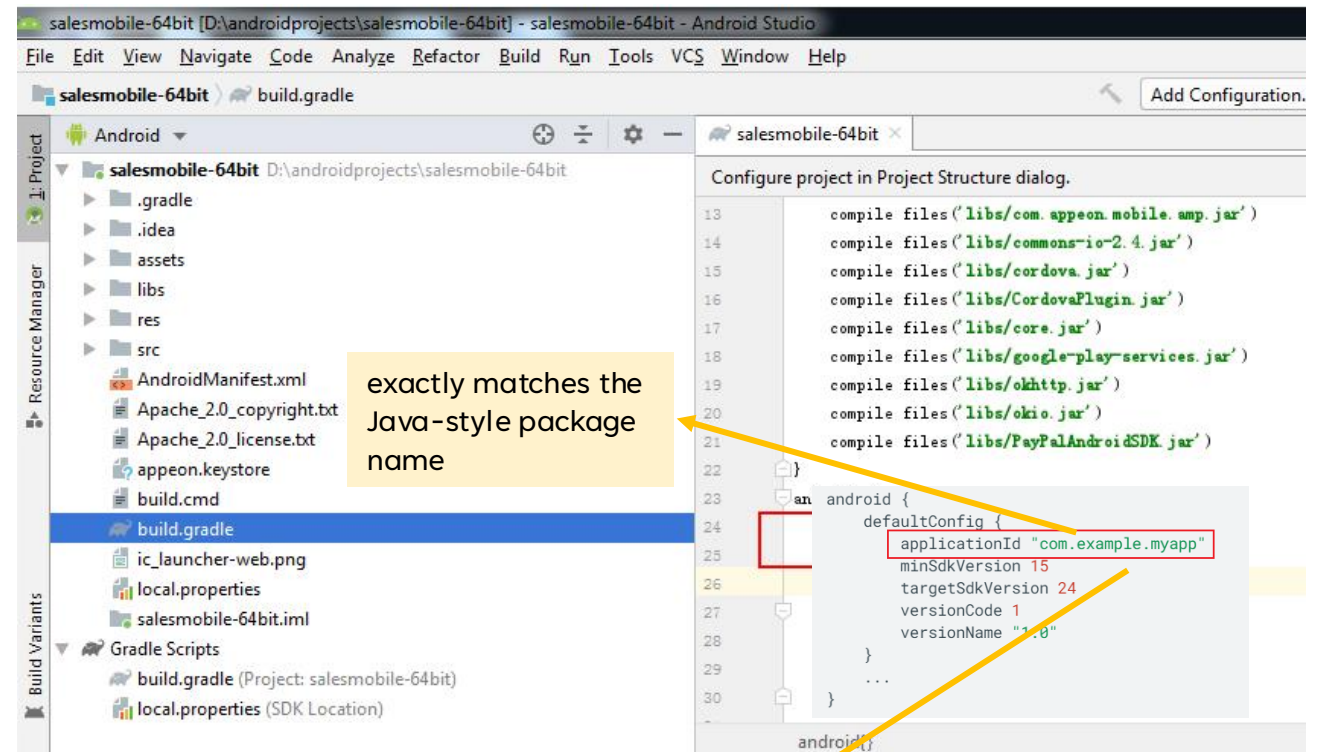
The image shows a terminal window with a red title bar. The prompt is `(kali㉿kali)-[~/pixel3]`. The command `ls -l 'Pixel 3/data/user_de/'` is entered and highlighted with a red box. The output shows two directories: `0` and `11`, both with permissions `drwxr-xr-x`, owned by `kali`, size `4096`, and date `Feb 15 2020`.

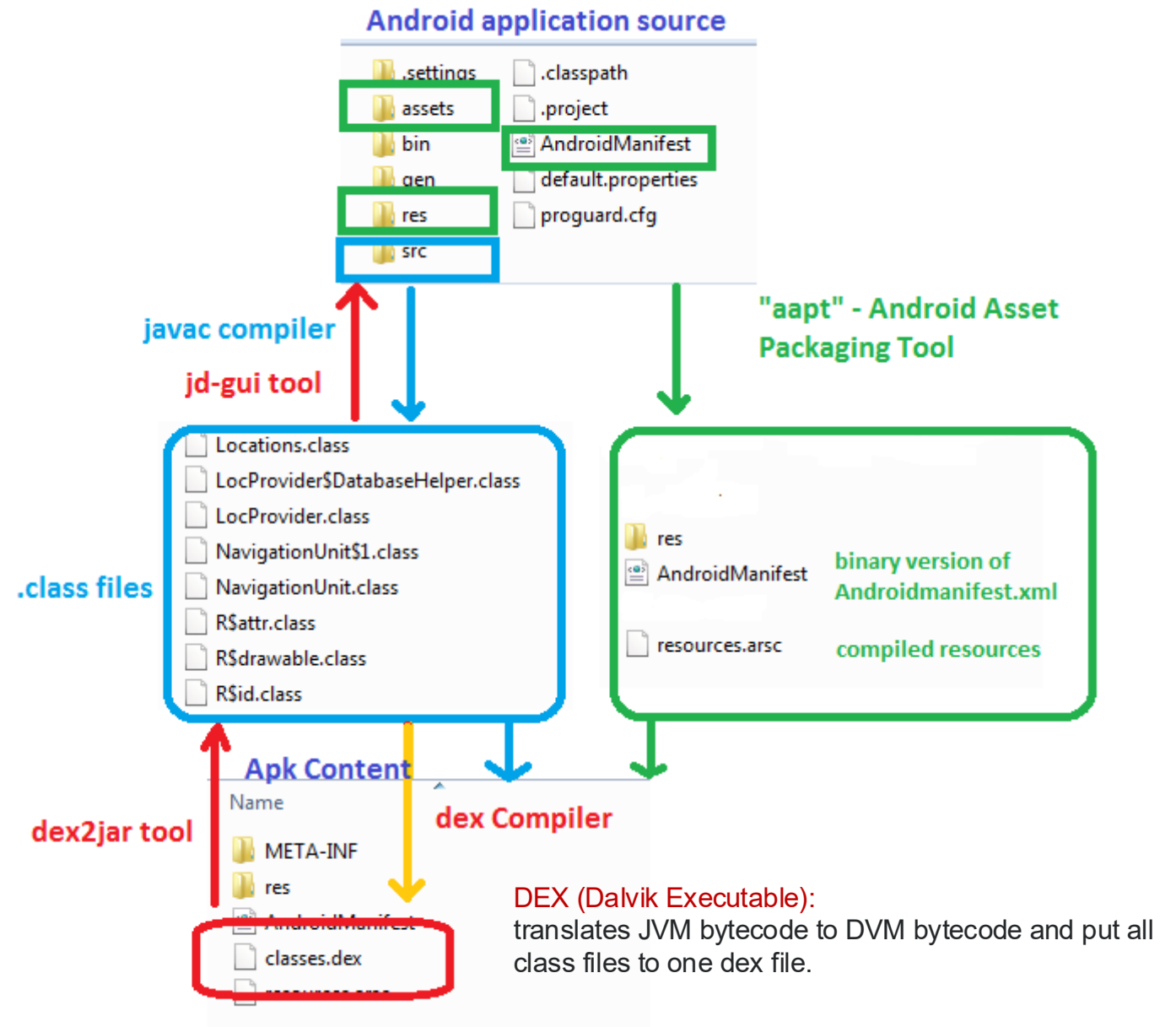
UNDERSTAND THE APPS/PACKAGE

Know where to extract forensic data

WHAT IS ANDROID (APPLICATION) PACKAGE .APK?

- Android Package (APK) is the package file format
 - used by the Android OS
 - zip + pre-defined specific structure
 - compiled to **.dex** format
- application ID is defined with the **applicationId** property in **build.gradle**
 - Use the ID to identify apps and the generated data





```
(kali㉿kali)-[~/pixel3]  
$ ls 'Pixel 3/data/data/'
```

Use *ApplicationID* as a folder to save application data

```
com.android.chrome  
com.android.hotwordenrollment.okgoogle  
com.android.hotwordenrollment.xgoogle  
com.android.keychain  
com.android.mtp  
com.android.nfc  
com.android.providers.calendar  
com.android.providers.contacts  
com.android.providers.downloads  
com.android.providers.media  
com.android.providers.telephony  
com.android.providers.userdictionary  
com.android.service.ims.presence  
com.android.traceur  
com.android.vending  
com.discord  
com.enflick.android.TextNow  
com.facebook.orca  
com.google.android.apps.chromecast.app  
com.google.android.apps.docs  
com.google.android.apps.docs.editors.docs  
com.google.android.apps.dreamliner  
com.google.android.apps.enterprise.dmagent  
com.google.android.apps.gcs  
com.google.android.apps.helprtc  
com.google.android.apps.magazines  
com.google.android.apps.maps  
com.google.android.apps.messaging  
com.google.android.apps.nexuslauncher  
com.google.android.apps.photos  
com.google.android.apps.pixelmigrate  
com.google.android.apps.safetyhub  
com.google.android.apps.tachyon  
com.google.android.apps.tips
```

AOSP Apps

GMS Apps

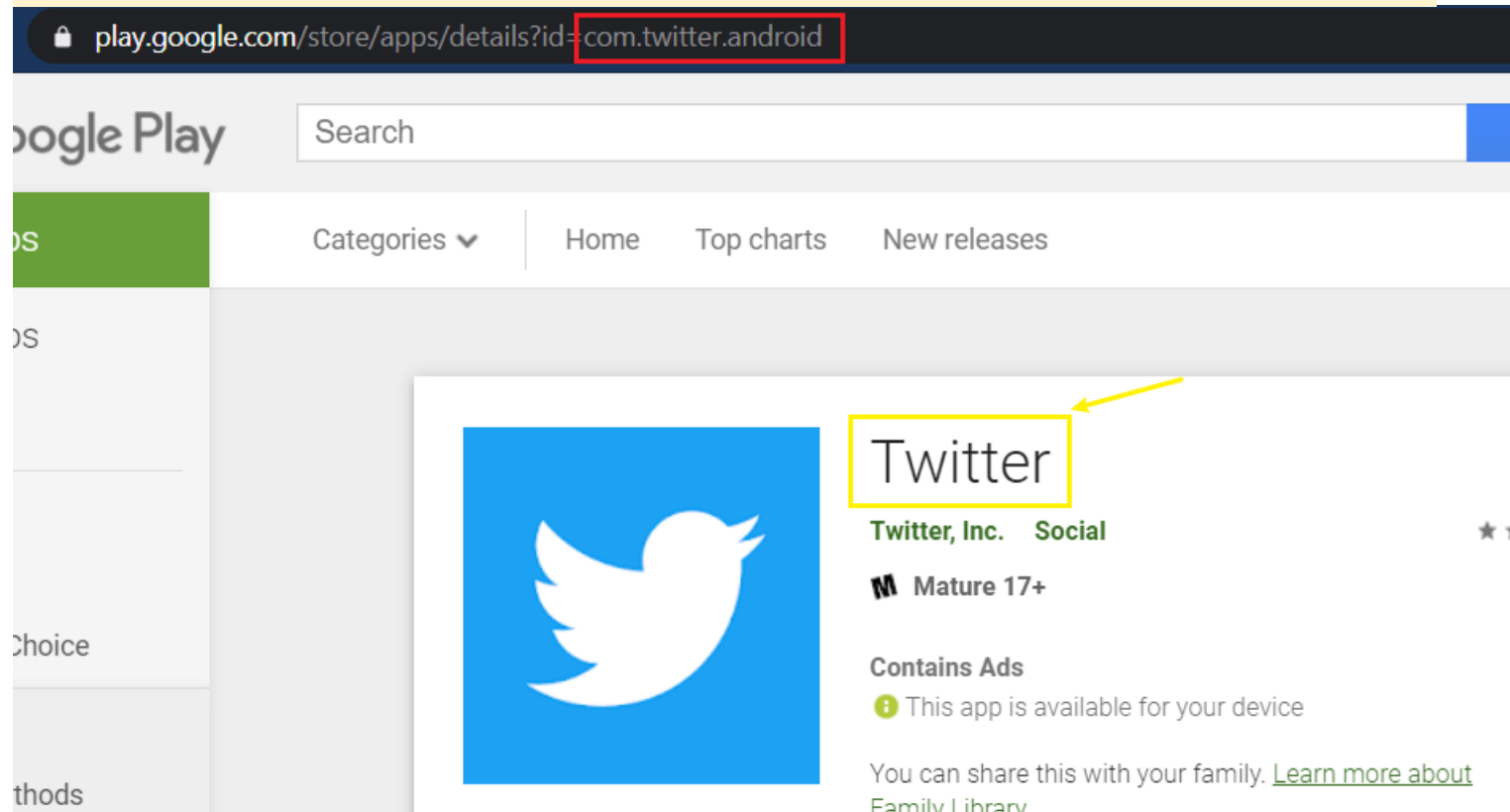
```
com.google.android.apps.turbo  
com.google.android.apps.tycho  
com.google.android.apps.walletnfcrel  
com.google.android.apps.wallpaper  
com.google.android.apps.wellbeing  
com.google.android.apps.work.oobconfig  
com.google.android.apps.youtube.music  
com.google.android.as  
com.google.android.calendar  
com.google.android.configupdater  
com.google.android.contacts  
com.google.android.deskclock  
com.google.android.dialer  
com.google.android.documentsui  
com.google.android.ext.services  
com.google.android.gm  
com.google.android.gms  
com.google.android.GoogleCamera  
com.google.android.googlequicksearchbox  
com.google.android.gsf  
com.google.android.ims  
com.google.android.inputmethod.latin  
com.google.android.keep  
com.google.android.markup  
com.google.android.music  
com.google.android.onetimeinitializer  
com.google.android.partnersetup  
com.google.android.pixel.setupwizard  
com.google.android.projection.gearhead  
com.google.android.settings.intelligence  
com.google.android.setupwizard  
com.google.android.syncadapters.contacts  
com.google.android.tts  
com.google.android.videos
```

```
com.google.android.webview  
com.google.android.wfcactivation  
com.google.android.youtube  
com.google.ar.core  
com.google.intelligence.sense  
com.google.vr.apps.ornament  
com.google.vr.vrcore  
com.imgur.mobile  
com.imo.android.imoous  
com.instagram.android  
com.instagram.threadsapp  
com.mewe  
com.mywickr.wickr2  
com.qti.ltebc  
com.radicalapps.cyberdust  
com.silentcircle.silentphone  
com.skout.android  
com.skype.raider  
com.snapchat.android  
com.spotify.music  
com.thinkyeah.galleryvault  
com.topjohnwu.magisk  
com.twitter.android  
com.venmo  
com.viber.voip  
com.whatsapp  
com.wire  
com.zhiliaoapp.musically  
jp.naver.line.android  
kik.android  
org.telegram.messenger  
org.thoughtcrime.securesms  
org.torproject.torbrowser
```

3rd Party

WHAT IS THE NAME OF APPLICATION FOR A GIVEN *APPLICATIONID*? (*TWITTER VS. COM.TWITTER.ANDROID*)

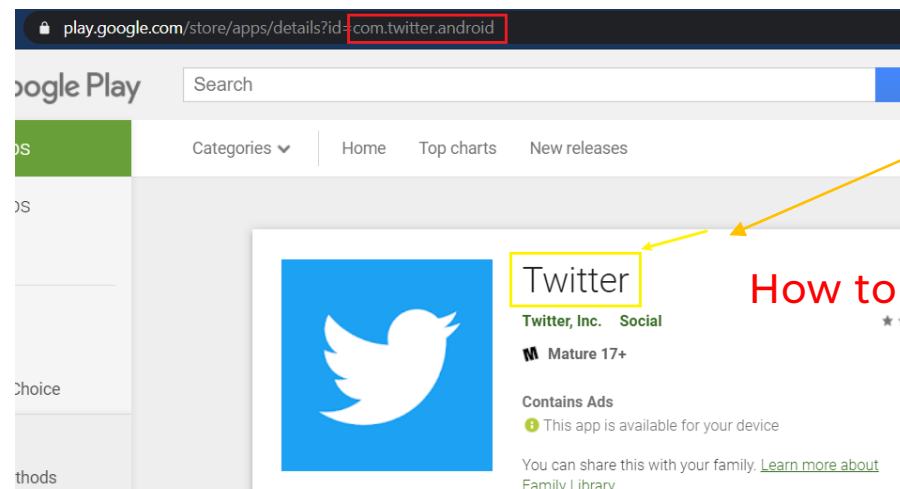
https://play.google.com/store/apps/details?id=com.twitter.android



HOW TO FIND THE NAME OF THE APPLICATION **AUTOMATICALLY?** (*COM.TWITTER.ANDROID*)

Steps

1. Send a http request to <https://play.google.com/store/apps/details?id=com.twitter.android>
2. Parse the html
3. Search for a string (name of the application) from a specific location



How to identify the location? (next slice)

Find inspect the Twitter name location in Chrome

The image illustrates the steps to inspect the Twitter name location in Chrome. On the left, a context menu is open over the Twitter logo, with the 'Inspect' option highlighted. An orange arrow points from 'Inspect' to the right, where the Chrome DevTools 'Elements' panel is shown. In the 'Elements' panel, the DOM tree is expanded to show the HTML structure, and the specific span element containing the text 'Twitter' is highlighted with a red box.

Twitter, Inc. Mature 17+ Contains Ads This app is You can share Family Library

Copy Ctrl+C
Search Google for "Twitter"
Print... Ctrl+P
Block element...
Google Translate
Mark the word "Twitter"
Save selection in Note Board
Inspect Ctrl+Shift+I

```
<div class="JNury Ekdnc">  
  <main class="LXrl4c">  
    <c-wiz jsrenderer="rDSaC" jsshadow jsdata="deferred-i7" data-p="%@.[\"com.twitter.android\",7],true,false]\" data-node-index="4;0\" jsmodel="hc6Ubd">  
      <c-wiz jsrenderer="uYq0pb" jsshadow jsdata="deferred-i8" data-p="%@.[\"com.twitter.android\",7],true]\" data-node-index="2;0\" jsmodel="hc6Ubd">  
        <div class="oQ6oV">  
          <div class="hkhL9e">...</div>  
          <div class="D0ZKYe">  
            <div class="rlnrKc">  
              <div class="sIskre">  
                <c-wiz jsrenderer="vVn0i" jsshadow jsdata="deferred-i9" data-p="%@.[\"com.twitter.android\",7]\" data-node-index="13;0\" jsmodel="hc6Ubd">  
                  <h1 class="AHFaub" itemprop="name">  
                    <span>Twitter</span> == $0  
                  </h1>  
                  <c-data id="i9" jsdata=" trIp0b;_;13 0KeYaf;_;1 0"></c-data>  
                </c-wiz>  
              </div>  
            </div>  
          </div>  
        </div>  
      </c-wiz>  
    </main>  
  </div>
```

Show the **location** of Twitter name

```
(kali㉿kali)-[~/pixel3]
$ curl --no-progress-meter https://play.google.com/store/apps/details?id=com.twitter.android |
grep -Pio 'itemprop="name"><span>Twitter</span>'
itemprop="name"><span>Twitter</span>
```

-P, --perl-regexp PATTERNS are Perl regular expressions
-o, --only-matching show only nonempty parts of lines that match
-i, --ignore-case ignore case distinctions in patterns and

Only show the name **Twitter** name (only match the word between
'itemprop="name"> and ****)

```
(kali㉿kali)-[~/pixel3]
$ curl --no-progress-meter https://play.google.com/store/apps/details?id=com.twitter.android |
grep -Pio 'itemprop="name"><span>\K.*?(?=</span>)'
Twitter ←
```

? : lazy match

\K : Keep text matched so far out of the return match

(?=pattern) : used as a zero-width look-ahead assertion after the text you want to output

LIST ALL APP NAMES AND PACKAGE ID

```
ls 'Pixel 3/data/data' \  
| while read package; do \  
curl --no-progress-meter \  
https://play.google.com/store/apps/details?id=$packa \  
ge \  
| grep -Pio 'itemprop="name"><span>\K.*?(?=</span>)' \  
&& echo $package; \  
done
```

```
(kali㉿kali)-[~/pixel3]  
$ ls 'Pixel 3/data/data' \  
| while read package; do \  
curl --no-progress-meter https://play.google.com/store/apps/details?id=$package \  
| grep -Pio 'itemprop="name"><span>\K.*?(?=</span>)' \  
&& echo $package; \  
done  
Google Chrome: Fast & Secure  
com.android.chrome  
Discord - Talk, Video Chat & Hang Out with Friends  
com.discord  
TextNow: Free Texting & Calling App  
com.enflick.android.TextNow  
Messenger - Text and Video Chat for Free  
com.facebook.orca  
Google Home  
com.google.android.apps.chromecast.app  
Google Drive  
com.google.android.apps.docs  
Google Docs  
com.google.android.apps.docs.editors.docs  
Google Apps Device Policy  
com.google.android.apps.enterprise.dmagent  
Google Support Services  
com.google.android.apps.helprtc  
Google News - Top world & local news headlines  
com.google.android.apps.magazines  
Google Maps - Navigate & Explore  
com.google.android.apps.maps  
Messages  
com.google.android.apps.messaging  
Pixel Launcher  
com.google.android.apps.nexuslauncher  
Google Photos  
com.google.android.apps.photos
```

ARE THERE ANY APPS WITH MONEY TRANSFER FUNCTION?

```
(kali㉿kali)-[~/pixel3]
$ ls 'Pixel 3/data/data' \
| while read package; do \
curl --no-progress-meter https://play.google.com/store/apps/details?id=$package \
| grep -Pio '<div jsname="sngebd">\K.*?(?=</div>)' | grep -io money\
&& echo $package; \
done
money
money
com.enflick.android.TextNow ←
MONEY
money
money
com.facebook.orca ←
money
com.google.android.apps.walletnfcrel ←
money
money
money
money
money
money
money
Money
Money
money
com.venmo ←
```

Interesting: It seems *venmo* mentioned money 10 times

PIXEL3 DEVICE INVESTIGATIONS

Hardware, Accounts, Mac address, Wi-Fi

OVERVIEW

- Device information
- Account information
- Mac address
- Wi-Fi

PIXEL 3 DEVICE INFORMATION

OVERVIEW

Description	Path
Build properties	Pixel 3/product/build.prop
	Pixel 3/system/build.prop
Carrier config	Pixel 3/data/user_de/0/com.android.providers.telephony/
Device serial number	Pixel 3/mnt/vendor
Bluetooth	Pixel 3/data/system/users/0/settings_secure.xml
Wifi	Pixel 3/data/misc/wifi/WifiConfigStore.xml

WHAT IS THE PRODUCT (PHONE) INFORMATION?

build.prop

- A system file that exists on every Android device.
- The file contains build information and other system properties which are used throughout the operating system.

```
(kali@kali)-[~/pixel3]
$ cat 'Pixel 3/product/build.prop'

# begin common build properties
# autogenerated by build/make/tools/buildinfo_common.sh
ro.product.build.date=Tue Dec 17 19:34:12 UTC 2019
ro.product.build.date.utc=1576611252
ro.product.build.fingerprint=google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys
ro.product.build.id=QQ1A.200105.003
ro.product.build.tags=release-keys
ro.product.build.type=user
ro.product.build.version.incremental=6079943
ro.product.build.version.release=10
ro.product.build.version.sdk=29
ro.product.product.brand=google
ro.product.product.device=blueline
ro.product.product.manufacturer=Google
ro.product.product.model=Pixel 3
ro.product.product.name=blueline
# end common build properties
```

When the current ROM was compiled

build information

software development kit: Version of the SDK used for android version

product information

Device's codename

https://source.android.com/security/bulletin/2020-01-01#2020-01-01-
details

source.android.com/security/bulletin/2020-01-01#2020-01-01-details

Set up Design **Secure** Develop Configure Reference Search

BULLETINS FEATURES TESTING BEST PRACTICES

Services [link](#), and is especially important for users who install apps from outside of Google Play.

/ Bulletins

2020-01-01 security patch level vulnerability details

In the sections below, we provide details for each of the security vulnerabilities that apply to the 2020-01-01 patch level. Vulnerabilities are grouped under the component they affect. There is a description of the issue and a table with the CVE, associated references, [type of vulnerability](#), [severity](#), and updated AOSP versions (where applicable). When available, we link the public change that addressed the issue to the bug ID, such as the AOSP change list. When multiple changes relate to a single bug, additional references are linked to numbers following the bug ID.

Framework

The most severe vulnerability in this section could enable a local malicious application to bypass user interaction requirements in order to gain access to additional permissions.

CVE	References	Type	Severity	Updated AOSP versions
CVE-2020-0001	A-140055304	EoP	Moderate	10
		EoP	High	8.0, 8.1, 9
CVE-2020-0003	A-140195904	EoP	High	8.0
CVE-2020-0004	A-120847476	DoS	High	8.0, 8.1, 9, 10

```
(kali@kali)-[~/pixel3]
$ grep patch 'Pixel 3/system/build.prop'
ro.build.version.security_patch=2020-01-01
```

DOES THE PIXEL HAVE A SD CARD?

```
kat@kat: ~/pixels
## Disable touch heatmap
#
ro.input.video_enabled=false
# end of device/google/crosshatch/product.prop
#
# ADDITIONAL PRODUCT PROPERTIES
#
ro.build.characteristics=nosdcard
persist.vendor.debug.sensors.accel_cal=1
masterclear.allow_retain_esim_profiles_after_fdr=true
ro.apk_verity.mode=1
ro.sys.sdcardfs=1
vendor.esel.loader_script_path=/sys/firmware/devicetree/b
th
ro.lmk.log_stats=true
ro.adb.secure=1
ro.lmk.low=1001
ro.lmk.medium=800
ro.lmk.critical=0
ro.lmk.critical_upgrade=false
```

WHAT IS THE MOBILE CARRIER/PHONE#?

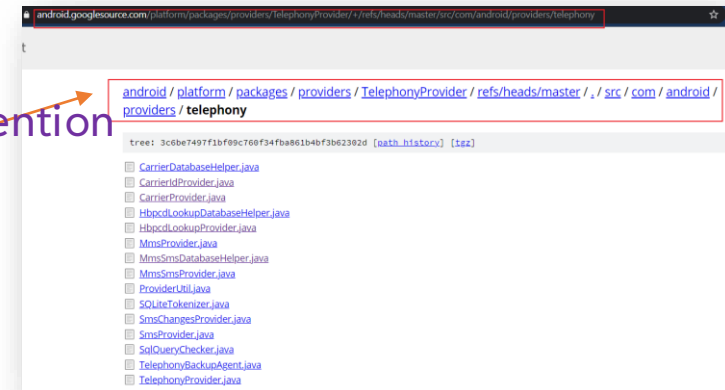
Exam all files under *com.android.providers.telephony* folder

naming convention

```
(kali@kali)-[~/pixel3]
$ tree 'Pixel 3/data/user_de/0/com.android.providers.telephony/'
Pixel 3/data/user_de/0/com.android.providers.telephony/
├── databases
│   ├── carrierIdentification.db
│   ├── CarrierInformation.db
│   ├── cellbroadcasts.db
│   ├── HbpcdLookup.db
│   ├── mmssms.db
│   └── telephony.db
└── shared_prefs
    ├── build-id.xml
    ├── CarrierIdProvider.xml
    ├── preferred-apn.xml
    └── preferred-full-apn.xml
```

all carrierIDs worldwide

QQ1A.200105.003



List tables on *telephony.db*

kali@kali: ~/pixel3 146x37

(kali@kali)-[~/pixel3]

```
$ sqlitebrowser 'Pixel 3/data/user_de/0/com.android.providers.telephony/databases/telephony.db'
```

DB Browser for SQLite - Pixel 3/data/user_de/0/com.android.providers.telephony/databases/telephony.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Open Project Save Project Attach Database Close Database

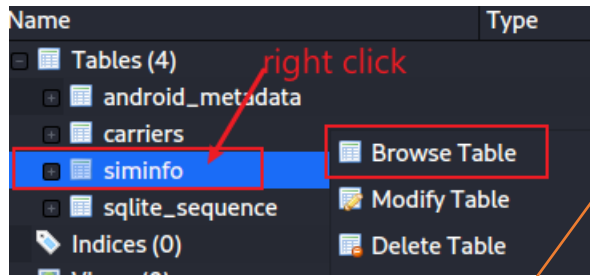
Database Structure Browse Data Edit Pragas Execute SQL

Create Table Create Index Modify Table Delete Table Print

Name	Type	Schema
Tables (4)		
android_metadata		CREATE TABLE android_metadata (locale TEXT)
carriers		CREATE TABLE carriers(_id INTEGER PRIMARY KEY,name TEXT DEFAULT '',numeric TEXT DEI
siminfo		CREATE TABLE siminfo(_id INTEGER PRIMARY KEY AUTOINCREMENT,icc_id TEXT NOT NULI
sqlite_sequence		CREATE TABLE sqlite_sequence(name,seq)

Edit Database C

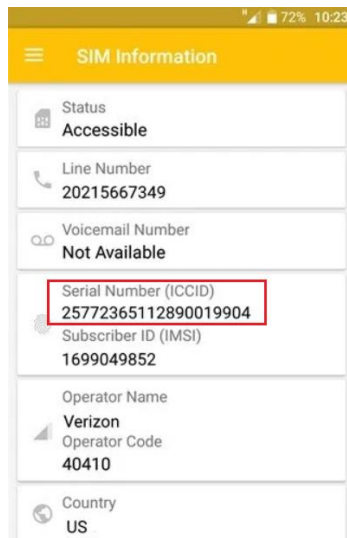
Mode: Text



- **ICCID** (Integrated Circuit Card Identifier) is a unique SIM card serial number.
- A full ICCID number is 19 or 20 characters.
- The number is printed on the back of your SIM (sometimes there is only the last 13 digits of ICCID)
 - you can also find it in your phone's settings. Read More about ICCID.
- Based on ICCID number you can check your SIM Country and International phone code.
 - <https://www.imei.info/iccid>

Table: **siminfo**

_id	icc_id	sim_id	display_name	carrier_name	e_so	color	number	ml	ro	mcc	mnc
1	8901260971148676693	-1	Google Fi		3	-1...	6513381146	1	0	310	260
2	89011203004056803842	0	Google Fi	Google Fi	3	-1...	+19195794674	1	0	310	120



mobile country code (MCC) and mobile network code (MNC) are used to find network providers

310	120	Sprint	Sprint Corporation	Operational	CDMA2000 1900 / LTE 850 / LTE 1900
310	260	T-Mobile	T-Mobile US	Operational	GSM 1900 / UMTS 1900 / UMTS 1700 / LTE 850 / LTE 700 / LTE 1900 / LTE 1700 / 5G 600

[https://en.wikipedia.org/wiki/Mobile_Network_Codes_in_ITU_region_3xx_\(North_America\)#United_States_of_America_%E2%80%93_US](https://en.wikipedia.org/wiki/Mobile_Network_Codes_in_ITU_region_3xx_(North_America)#United_States_of_America_%E2%80%93_US)

Check *carrierconfig* information to verify the carriers

```
kali@kali: ~/pixel3 102x37
(kali@kali)-[~/pixel3]
$ tree 'Pixel 3/data/user_de/0/com.android.phone/'
Pixel 3/data/user_de/0/com.android.phone/
├── files
│   ├── carrierconfig-com.google.android.apps.tycho-89011203004056803842-10015.xml
│   ├── carrierconfig-com.google.android.carrier-89011203004056803842-10015.xml
│   └── carrierconfig-com.google.android.carrier-8901260971148676693-10014.xml
└── shared_prefs
    ├── com.android.phone_preferences.xml
    └── _has_set_default_values.xml

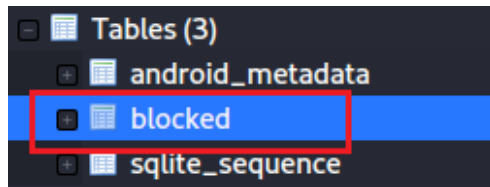
2 directories, 5 files
```

Java naming convention

WHAT IS BLOCKED PHONE NUMBER?

Show blockednumber

```
(kali@kali)-[~/pixel3]  
$ sqlitebrowser 'Pixel 3/data/user_de/0/com.android.providers.blockednumber/databases/blockednumbers.db'
```



Database Structure Browse Data Edit Pragma Execute SQL

Table: blocked

	_id	original_number	e164_number
	Filter	Filter	Filter
1	1	+19195121037	+19195121037

ICCID id

```
(kali㉿kali)-[~/pixel3]
$ cat 'Pixel 3/data/user_de/0/com.android.phone/files/carrierconfig-com.google.android.carrier-89011203004056803842-10015.xml'
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<carrier_config>
<package_version>8</package_version>
<bundle_data>
<string name="gps.lpp_profile">2</string>
<string-array name="enable_apps_string_array" num="0" />
<boolean name="carrier_app_required_during_setup_bool" value="true" />
<boolean name="config_telephony_use_own_number_for_voicemail_bool" value="true" />
<string name="carrier_config_version_string">fi_us-12000000333.16</string>
<string name="carrier_name_string">Google Fi</string>
<boolean name="carrier_name_override_bool" value="true" />
</bundle_data>
</carrier_config>
```

ICCID id

```
(kali㉿kali)-[~/pixel3]
$ cat 'Pixel 3/data/user_de/0/com.android.phone/files/carrierconfig-com.google.android.carrier-8901260971148676693-10014.xml'
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<carrier_config>
<package_version>8</package_version>
<bundle_data>
<string name="gps.lpp_profile">2</string>
<string-array name="enable_apps_string_array" num="0" />
<boolean name="carrier_app_required_during_setup_bool" value="true" />
<boolean name="config_telephony_use_own_number_for_voicemail_bool" value="true" />
<string name="carrier_config_version_string">fi_us-12000000333.16</string>
<string name="carrier_name_string">Google Fi</string>
<boolean name="carrier_name_override_bool" value="true" />
</bundle_data>
</carrier_config>
```

WHAT IS SERIAL NUMBER OF THE DEVICE?

- Hardware configurations may contain the serial number of the device
 - e.g., camera, audio devices configs
- Find vendor specific configurations associated with hardware devices
- Where are these configuration files?

```
(kali@kali)-[~/pixel3]
$ tree 'Pixel 3/mnt/vendor/' -L 2
Pixel 3/mnt/vendor/
├── persist
│   ├── audio
│   ├── battery
│   ├── camera
│   ├── data
│   ├── display
│   ├── haptics
│   ├── rfs
│   ├── secure_element
│   ├── sensors
│   └── time
```

Search key word “serial number” under the */vendor* folder

```
kali@kali: ~/pixel3 146x37
(kali@kali)-[~/pixel3]
$ grep -ri 'serial number' 'Pixel 3/mnt/vendor/'
Pixel 3/mnt/vendor/persist/sensors/tmd2725_als_factory.txt:Serial Number: 8CEX1N716

(kali@kali)-[~/pixel3]
$ grep -ri 'EX1N716' 'Pixel 3'
Pixel 3/mnt/vendor/persist/sensors/edgesense_roller_cal/4_results/measurements.csv:SYS_SN,8CEX1N716
Pixel 3/mnt/vendor/persist/sensors/tmd2725_als_factory.txt:Serial Number: 8CEX1N716
Pixel 3/mnt/vendor/persist/camera/front_stereo.cal:      "id": "8CEX1N716",
Pixel 3/mnt/vendor/persist/camera/front_stereo.cal:      "debug_path": "/home/innorev/camera_clifford/destination/output/8CEX1N716-1544779510.32"
Pixel 3/mnt/vendor/persist/camera/front_stereo.cal:      "/home/innorev/camera_clifford/destination/8CEX1N716",
Pixel 3/mnt/vendor/persist/camera/front_stereo.cal:      "8CEX1N716",
```

Explanation of **grep -ri**:

grep: A command-line utility used to search for text patterns within files.

-r: Recursively searches all files in the specified directory and its subdirectories.

-i: Makes the search case-insensitive, meaning it will match both uppercase and lowercase

`grep -ri "search_term" /path/to/directory`

WHAT IS BLUETOOTH MAC ADDRESS?

```
kali@kali: ~/pixel3 82x21

(kali@kali)-[~/pixel3]
$ grep -i 'bluetooth' 'Pixel 3/data/system/users/0/settings_secure.xml'
<setting id="65" name="bluetooth_name" value="Pixel 3" package="android" default
Value="Pixel 3" defaultSysSet="true" />
<setting id="71" name="bluetooth_address" value="7C:D9:5C:AC:A2:CE" package="and
roid" defaultValue="7C:D9:5C:AC:A2:CE" defaultSysSet="true" />
```

WHAT IS WI-FI SERVICE SET IDENTIFIER (SSID)? (*WIFICONFIGSTORE.XML*)

```
(kali㉿kali)-[~/pixel3]
$ cat 'Pixel 3/data/misc/wifi/WifiConfigStore.xml'
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<WifiConfigStoreData>
<int name="Version" value="2" />
<Integrity>
<byte-array name="EncryptedData" num="48">e71394aa95f2d767a1c46e787ca99c0478e12
e8a4082529d69a15d5bc49f28ca21a54</byte-array>
<byte-array name="IV" num="12">18234dad790da5cb3ba8eceb</byte-array>
</Integrity>
<NetworkList />
<MacAddressMap>
<map name="MacMapEntry">
<string name="&quot;CcookiesDcastleR5&quot;;WPA_PSK">ae:f6:c7:a8:a5:c2</string>
</map>
</MacAddressMap>
<PasspointConfigData>
<long name="ProviderIndex" value="0" />
</PasspointConfigData>
</WifiConfigStoreData>
```

SSID

a router's MAC

- For old version/other brand of Android
 - */data/misc/wifi/wpa_supplicant.conf* (possible)

```
Example NOT from Pixel 3
# more wpa_supplicant.conf
ctrl_interface=eth0
update_config=1

network={
    ssid="SomeWifiNetwork"
    psk="MyPlaintextPassword"
    key_mgmt=WPA-PSK
    priority=1
}
#
```

ACCOUNT INFORMATION

OVERVIEW

Users' ID, owner	Pixel 3/data/system/users/userlist.xml
User name,	Pixel 3/data/system/users/0.xml Pixel 3/data/system/users/1.xml
Gmail account	Pixel 3/data/data/com.google.android.gm/shared_prefs/Gmail.xml Pixel 3/data/data/com.google.android.gm/shared_prefs/UnifiedEmail.xml
Gmail with play store	Pixel 3/data/data/com.android.vending/cache/logs/
App account and password	Pixel 3/data/system_ce/0/account_ce.db Pixel 3/data/system_de/0/account_de.db

0 - is a device owner/primary user

TWO

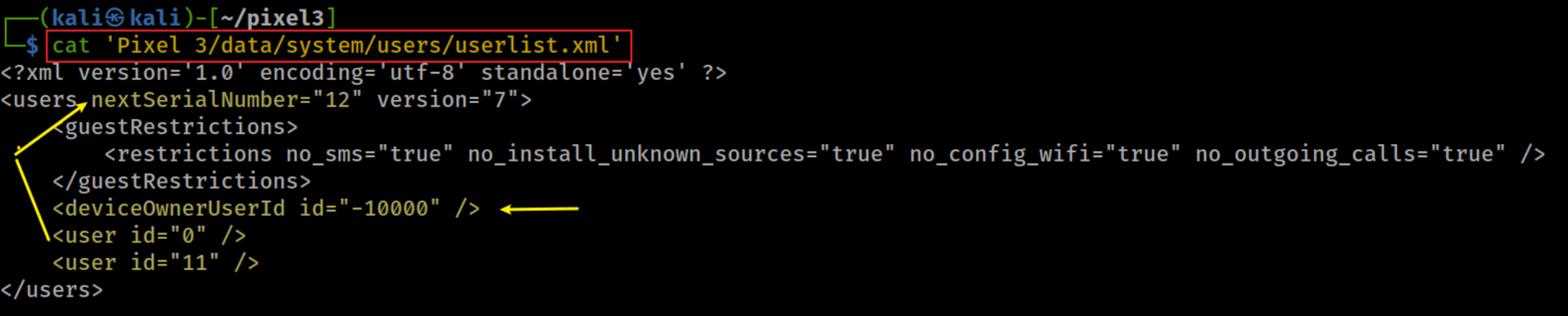
```
(kali@kali)-[~/pixel3]
$ tree 'Pixel 3/data/system/users'
Pixel 3/data/system/users
├── 0
│   ├── app_idle_stats.xml
│   ├── appwidgets.xml
│   ├── package-restrictions.xml
│   ├── photo.png
│   ├── registered_services
│   │   ├── android.accounts.AccountAuthenticator.xml
│   │   └── android.content.SyncAdapter.xml
│   ├── roles.xml
│   ├── runtime-permissions.xml
│   ├── settings_config.xml
│   ├── settings_global.xml
│   ├── settings_secure.xml
│   ├── settings_ssaid.xml
│   ├── settings_system.xml
│   ├── wallpaper
│   ├── wallpaper_info.xml
│   └── wallpaper_orig
├── 0.xml
├── 11
│   ├── app_idle_stats.xml
│   ├── appwidgets.xml
│   ├── device_policies.xml
│   ├── package-restrictions.xml
│   ├── photo.png
│   ├── registered_services
│   │   ├── android.accounts.AccountAuthenticator.xml
│   │   └── android.content.SyncAdapter.xml
│   ├── roles.xml
│   ├── runtime-permissions.xml
│   ├── settings_secure.xml
│   ├── settings_ssaid.xml
│   ├── settings_system.xml
│   ├── wallpaper
│   ├── wallpaper_info.xml
│   └── wallpaper_orig
├── 11.xml
└── userlist.xml
```

used for multi-user support.

keeps info on all users in the system

WHO IS THE OWNER OF THE DEVICE?

```
(kali㉿kali)-[~/pixel3]
$ cat 'Pixel 3/data/system/users/userlist.xml'
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<users nextSerialNumber="12" version="7">
  <guestRestrictions>
    <restrictions no_sms="true" no_install_unknown_sources="true" no_config_wifi="true" no_outgoing_calls="true" />
  </guestRestrictions>
  <deviceOwnerUserId id="-10000" />
  <user id="0" />
  <user id="11" />
</users>
```



WHEN DID THE DEVICE OWNER LAST TIME LOG IN?

```
(kali㉿kali)-[~/pixel3]
$ date -d @1581695517 -u
Fri 14 Feb 2020 03:51:57 PM UTC
```

```
(kali㉿kali)-[~/pixel3]
$ cat 'Pixel 3/data/system/users/0.xml'
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="0" serialNumber="0" flags="19" created="0" lastLoggedIn="1581695517124" lastLoggedInFingerprint="google/blueline/blueline:10/QQ1A.2
00105.003/6079943:user/release-keys" icon="/data/system/users/0/photo.png" profileBadge="0">
  <name>This Is</name>
  <restrictions />
</user>
```

7 ⚙

blueline: Android

```
(kali㉿kali)-[~/pixel3]
$ cat 'Pixel 3/data/system/users/11.xml'
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="11" serialNumber="11" flags="16" created="1581644358241" lastLoggedIn="1581684601922" lastLoggedInFingerprint="google/blueline/blue
line:10/QQ1A.200105.003/6079943:user/release-keys" icon="/data/system/users/11/photo.png" profileBadge="0">
  <name>User 2</name>
  <restrictions />
</user>
```

7 ⚙

- E: Extended Regular Expression
- i: ignore case
- o: show only match, not whole lines
- r: recursive
- h: don't show file name
- l: ignore binary files

list top 20 email

```
kali@kali: ~/pixel3 88x30
(kali@kali)-[~/pixel3]
$ grep -EiorhI '([[:alnum:]]_.-]+@[[:alnum:]]_.-]+?\.[[:alpha:]]{2,6})' 'Pixel 3' | sort
| uniq -c | sort -nr | head -n 20
608 thisisdfir@gmail.com
176 extension-devs@eff.org
120 thisisdfirtwo@gmail.com
96 19842352054_1oHEUWGxDxo499MnzUJStpXDQ1YoMyeXX406Faqv4Acw@prod.tncp.textnow.com
40 android.hardware.graphics.mapper@2.1.so
40 android.hardware.graphics.mapper@2.0.so
40 android.hardware.graphics.common@1.1.so
40 android.hardware.graphics.common@1.0.so
38 support@enflick.com
38 android.hardware.renderscript@1.0.so
38 19842032223@prod.tncp.textnow.com
32 ff7a7f86a2e8481abacfb5e046d5046a@patrol-sg.snssdk.com
24 android.hidl.safe_union@1.0.so
24 android.hidl.memory.token@1.0.so
24 android.hidl.memory@1.0.so
24 android.hardware.graphics.mapper@3.0.so
24 android.hardware.graphics.common@1.2.so
20 libstagefright_bufferpool@2.0.1.so
18 android.system.suspend@1.0.so
18 android.hidl.token@1.0-utils.so
```

Count all emails

```
(kali@kali)-[~/pixel3]
$ grep -EiorhI '([[:alnum:]]_.-]+@[[:alnum:]]_.-]+?\.[[:alpha:]]{2,6})' 'Pixel 3' | sort
| uniq -c | sort -nr | wc -l
618
```

WHAT IS THE APP ID OF GMAIL?

- Every Android app has a unique application ID
 - looks like a Java package name, such as *com.example.myapp*
- Android app ID never changes
 - If upload a new version of app, the application ID (and the certificate you sign it with) must be the same as the original APK
 - if you change the application ID, Google Play Store treats the APK as a completely different app.
- Good for forensic analysis
 - Easy to identify apps in a device



Google Play

Search



Apps

Categories ▾

Home

Top charts

New releases

My apps

Shop



Games

Kids

Editors' Choice

Account

Payment methods

Play Points

New

Every app has an ID/package name ID



Gmail

Google LLC Communication

★★★★☆ 8,601,956

Everyone

Contains Ads

This app is available for your device

You can share this with your family. [Learn more about Family Library.](#)

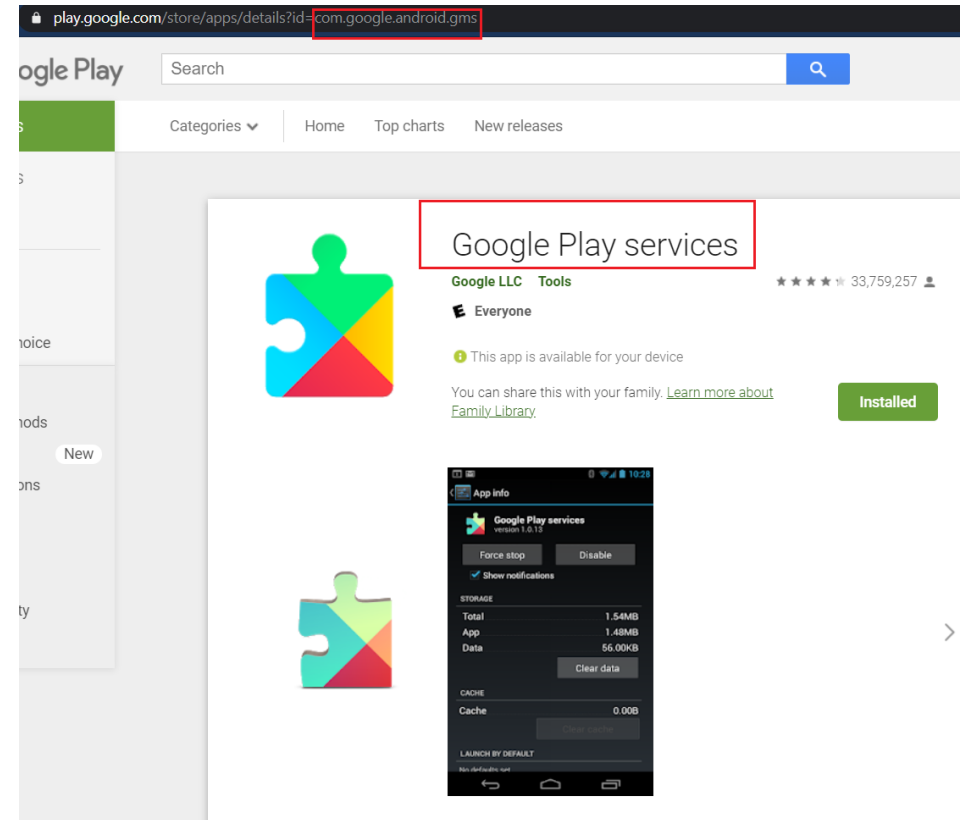
Installed

WHAT IS THE APP GOOGLE PLAY SERVICES ID?

- Google Play Services helps integrating Google's advanced functionalities with other applications.
 - It is one of the most important parts of Android
- Core functionalities
 - Authentication to your Google service
 - Synchronized contacts,
 - Access to all the latest user privacy settings, and
 - Higher quality, lower-powered location-based services, maps
 - Offline search

<https://play.google.com/store/apps/details?id=com.google.android.gms>

ms



don't /can't delete it!

Google Maps API: This API allows other applications to access Google Maps without leaving the app. If you have used an app to book a cab or navigate to a booked hotel, then you must have used this service.

Location APIs: Location API can be used to perform automatic actions when a user enters or leaves a specific geographic boundary. Fused Location provider means to reduce power usage while using location services.

Google Drive API: If you have Google Drive installed on your device, sharing and storing files from other apps is made easier with this API.

Google Play Game Services: These services are used to create a more competitive and immersive experience in multiplayer gaming sessions through leaderboards and achievements. Moreover, Saved Games API ensures that you do not lose your progress in a game by syncing and saving your game data.

Google+ platform: You must have come across apps that provide you with “Sign-in with Google+” login option. This service allows you to use a single username and password to access multiple accounts on multiple apps.

Google Cast Android API: It allows Android applications to cast content on-screen to other devices via Google Cast.

Google Mobile Ads: Google Mobile Ads is used to integrate advertisements into applications and display intelligent ads such as those based on location or browsing history. This is how thousands of app developers monetize their app.

Other service: Google Wallet, Google Play Protect, Google Fit API, Google account authentication methods

FIND ALL GMAIL ADDRESSES APPEARED IN THE PHONE?

```
kali@kali: ~/pixel3 89x30
(kali@kali)-[~/pixel3]
$ grep -EiorhI '([[:alnum:]]_.-]+@gmail\.com)' 'Pixel 3' | sort | uniq -c | sort -nr | head

608 thisisdfir@gmail.com
120 thisisdfirtwo@gmail.com
10 nc_1_mail_thisisdfir@gmail.com
6 feature_list_thisisdfir@gmail.com
4 ncg_1_account_thisisdfirtwo@gmail.com
4 ncg_1_account_thisisdfir@gmail.com
4 flagsLastLogged-thisisdfir@gmail.com
2 waa-enabledthisisdfir@gmail.com
2 user-settings-cachethisidfir@gmail.com
2 toc-cookiethisidfir@gmail.com
```

WHICH GMAIL IS REGISTERED FOR GOOGLE SERVICE FOR PIXEL 3?

- Google use Gmail credential as a Single Sign-On (SSO)
 - for all the service they provide
- Search for generated data by Gmail application
 - Gmail package *com.google.android.gm*

Exam all generated data/config of Gmail

```
(kali@kali)-[~/pixel3]
$ tree 'Pixel 3/data/data/com.google.android.gm' -L 1
Pixel 3/data/data/com.google.android.gm
├── app_webview
├── cache
├── databases
├── files
├── no_backup
└── shared_prefs
```

Gmail package

perference often is associated with gmail info

Show shared preference/configurations

```
(kali@kali)-[~/pixel3]
$ tree 'Pixel 3/data/data/com.google.android.gms/shared_prefs' -L 1
Pixel 3/data/data/com.google.android.gms/shared_prefs
├── ACCOUNT_COUNT_MAP.xml
├── Account-thisisdfir@gmail.com.xml
├── AndroidMail.Main.xml
├── ANIMATABLE_LOGO_VIEW_FILE_KEY.xml
├── attachmentManager_-1294433372.xml
├── c9edm_-1294433372.xml
├── com.google.android.apps.gmail.accountDumpState&#58;thisisdfir@gmail.com&#58;com.google.xml
├── com.google.android.apps.gmail.dumpState.xml
├── com.google.android.apps.gmail.notifications&#58;thisisdfir@gmail.com.xml
├── com.google.android.gms_preferences.xml
├── com.google.android.gms.appid.xml
├── COMPOSE_UPLOADERS.xml
├── FirebaseAppHeartBeat.xml
├── FlagPrefs.xml
├── G6yPrefs.xml
├── GigSyncEngine_thisisdfir@gmail.com.xml
├── GMAIL_NETWORK_LOGGING_DEPOT.xml
├── Gmail.xml ←
├── _has_set_default_values.xml
├── MailAppProvider.xml
├── phenotype__com.google.android.libraries.social.populous.xml
├── timeZoneDbVersion.xml
├── triaged_notification_receive_times_thisisdfir@gmail.com.xml
├── UnifiedEmail.xml ←
├── warm_accounts_preferences.xml
├── watermark_notification_tokens_thisisdfir@gmail.com.xml
└── WebViewChromiumPrefs.xml
```

Check Gmail preference setting

```
(kali㉿kali)-[~/pixel3]
$ cat 'Pixel 3/data/data/com.google.android.gms/shared_prefs/Gmail.xml'
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
  <string name="thisisdfr-account-alias">thisisdfr</string>
  <string name="858233690-account-alias">858233690</string>
  <string name="TextNow-account-alias">TextNow</string>
  <string name="active-account">thisisdfr@gmail.com</string>
  <string name="removal-action">archive</string>
  <boolean name="conversation-overview-mode" value="true" />
  <string name="Signal-account-alias">Signal</string>
  <boolean name="conversation-list-swipe" value="true" />
  <boolean name="force_show_welcome_tour" value="false" />
  <boolean name="confirm-archive" value="false" />
  <boolean name="confirm-send" value="false" />
  <string name="thisisdfr@gmail.com-account-alias">thisisdfr@gmail.com</string>
  <string name="Skype-account-alias">Skype</string>
  <string name="+19195794674-account-alias">+19195794674</string>
  <string name="WhatsApp-account-alias">WhatsApp</string>
  <boolean name="confirm-delete" value="false" />
  <int name="welcome_tour_version" value="1" />
  <boolean name="mail-enable-threading" value="true" />
  <string name="cache-google-accounts-synced">thisisdfr@gmail.com</string>
</map>
```

Gmail is associated with

- phone
- apps: Textnow, Signal, Skyper, whatapp

(kali@kali)-[~/pixel3]

\$ cat 'Pixel 3/data/data/com.google.android.gm/shared_prefs/UnifiedEmail.xml'

```
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
```

```
<map>
```

```
  <int name="offline-search-index-corpus-version" value="2" />
```

```
  <boolean name="email-accounts-import-tried" value="true" />
```

```
  <long name="analytics-send-nb_accounts-epoch" value="1581644062878" />
```

```
  <int name="auto-advance-mode" value="0" />
```

```
  <int name="offline-search-index-schema-version" value="2" />
```

```
  <set name="display_images" />
```

```
  <boolean name="conversation-photo-teaser-shown-three" value="true" />
```

```
  <boolean name="sc-onboarding-shown-v2" value="true" />
```

```
  <boolean name="confirm-archive" value="false" />
```

```
  <string name="gm-density">DEFAULT</string>
```

```
  <long name="dots_animation_timestamp" value="1581643980962" />
```

```
  <long name="owners_name_cache_updated_timestamp" value="1581643980626" />
```

```
  <string name="email_address_prefix:thisisdfr@gmail.com">This Is DFIR</string>
```

```
  <string name="swipe-left-action">archive</string>
```

WHICH GMAIL IS REGISTERED IN PLAY STORE?

Check Gmail information for Play store cache

```
(kali㉿kali)-[~/pixel3]
$ tree '/home/kali/pixel3/Pixel 3/data/data/com.android.vending/cache/logs/'
/home/kali/pixel3/Pixel 3/data/data/com.android.vending/cache/logs/
├── com.google.thisisdfir%40gmail.com.metalog
│   ├── play_metalog.log
│   └── null_account.metalog
├── logs_upload_attempt.log
└── play_metalog.log
```

gmail.com cached for download Apps

Google Play store

We can't find/don't know anything from *play_metalog.log*

```
(kali㉿kali)-[~/pixel3]
$ xxd 'Pixel 3/data/data/com.android.vending/cache/logs/com.google.thisisdfir%40gmail.com.metalog/play_metalog.log'
00000000: 08f4 c9de b084 2e60 da46 6801 .....`.Fh. ←
```

WHICH GMAIL USED FOR SCHEDULING JOBS?

- A Job in Android is a background tasks, e.g.,
 - receive a message
 - read data from a database
- Android provides a *JobScheduler*
 - It allows you to specify conditions for running work
 - e.g., a big downloading job is to run when the device is plugged in and has a WIFI connection.
- *Job-Info* is stored in */data/system/job/jobs.xml*
 - *Job-info* has a list of *job* objects
 - Each *job* contains forensic information

JOB OBJECT INFORMATION

* all time related parameters are specified in
milliseconds

- make the job periodic (a repeating job within a given period)
- the interval between periodic job runs
- the maximum execution delay – how long to wait to kick off the job (not a parameter for a periodic job task)
- the minimum amount of time delay before starting the job (not a parameter for a periodic job task)
- the device must be plugged in and charging in order to run the job
- the device must be idle (“in an idle maintenance window”) in order to run the job
- the device must have a network connection (either WIFI, unmetered, or any)
- the device must not be connected to a network
- the job should be rescheduled if the device reboots (called a “persistent” job).
- Extra: account name/Gmail maybe associated with the job

Time

86400000 = 24

Millisecond Hour

Formula divide the time value by 3.6e+6

Convert epoch to human-readable date and vice versa

1581771242040

Timestamp to Human date [batch convert]

Supports Unix timestamps in seconds, milliseconds, microseconds and nanoseconds.

Assuming that this timestamp is in **milliseconds**:

GMT : Saturday, February 15, 2020 12:54:02.040 PM

Your time zone : Saturday, February 15, 2020 7:54:02.040 AM GMT-05:00

Relative : A year ago

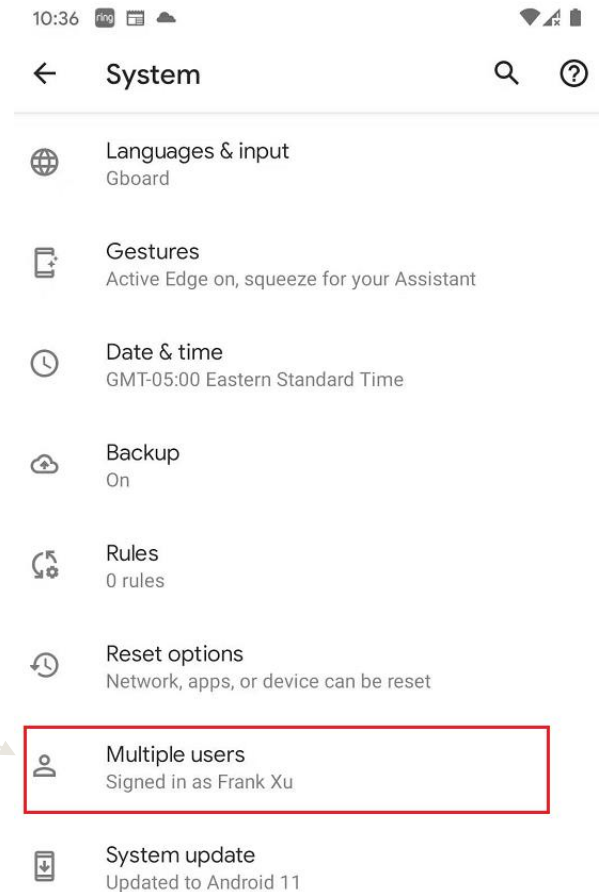
```
</job>
<job jobId="100567" package="android" class="com.android.server.content.SyncJobService" sourcePackageName="com.google.android.calendar">
  <constraints net-capabilities="94208" net-unwanted-capabilities="0" net-transport-types="0" />
  <periodic period="86400000" flex="4320000" deadline="1581771242040" delay="1581766922040" />
  <extras>
    <string name="provider">com.android.calendar</string>
    <int name="reason" value="-4" />
    <int name="source" value="4" />
    <int name="userId" value="11" />
    <long name="expectedRuntime" value="0" />
    <int name="syncExemptionFlag" value="0" />
    <string name="owningPackage">com.google.android.calendar</string>
    <boolean name="SyncManagerJob" value="true" />
    <boolean name="isPeriodic" value="true" />
    <boolean name="allowParallelSyncs" value="false" />
    <long name="periodMillis" value="86400000" />
    <int name="sourcePeriodicId" value="-1" />
    <int name="jobId" value="100567" />
    <int name="owningUid" value="1110178" />
    <long name="flexMillis" value="3456000" />
    <string name="accountName">thisisdirtwo@gmail.com</string>
    <string name="accountType">com.google</string>
    <int name="retries" value="0" />
    <pbundle_as_map name="syncExtras" />
  </extras>
</job>
```

a calendar activities is running on background. The activity is associated with **thisisdirtwo@gmail.com**

Extra:

If your job needs some information from your app to perform its work, you can pass along primitive data types as extras in the JobInfo.

Android 10 allows to add multiple users to share the phone space, customized screens, apps. That is the reason you may see multiple Gmail associated with jobs



Find two emails that are associated with **scheduled jobs**

```
kali@kali: ~/pixel3 90x24

(kali@kali)-[~/pixel3]
$ grep -Eio '([[:alnum:]]_[-]+@[[:alnum:]]_[-]+\.[[:alpha:]]{2,6})' 'Pixel 3/data/system/job/jobs.xml' | sort | uniq -c
43 thisisdfr@gmail.com ←
36 thisisdfrtwo@gmail.com ←
```

LIST POSSIBLE APPS USERNAME AND PASSWORD (*ACCOUNT_CE.DB/ACCCO*)

DB Browser for SQLite - /home/kali/pixel3/Pixel 3/data/system_ce/0/accounts_ce.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Open Project Save Project Attach

Database Structure Browse Data Edit Pragma Execute SQL

Table: accounts

	_id	name	type	password
1	1	thisisdfr@gmail.com	com.google	aas_et/...
2	2	858233690	org.telegram.messenger	
3	3	Signal	org.thoughtcrime.securesms	NULL
4	4	imo HD	com.imo.android.imo5	NULL
5	5	TikTok	com.zhiliaoapp.musically	NULL
6	6	Messenger	com.facebook.messenger	NULL
7	7	TDfir	com.twitter.android.auth.login	NULL
8	8	WhatsApp	com.whatsapp	NULL
9	10	thisisdfr	com.silentcircle.account	dummyPassword
10	11	Skype	com.skype.raider	NULL
11	12	TextNow	com.enflick.android.TextNow.account	
12	13	+19195794674	com.viber.voip	961ea18e43b4a2b35216e208d9...
13	14	Duo	com.google.android.apps.tachyon	NULL

```
(kali@kali)-[~/pixel3]
$ tree -L 1 'Pixel 3/data/system_ce/0/'
Pixel 3/data/system_ce/0/
├── accounts_ce.db
├── launch_params
├── recent_images
├── recent_tasks
├── shortcut_service
└── snapshots
```

LIST POSSIBLE APPS USERNAME AND PASSWORD (*ACCOUNT_DE.DB/ACC*)

```
(kali@kali)-[~/pixel3]
$ tree -L 1 'Pixel 3/data/system_de/0/'
Pixel 3/data/system_de/0/
├── accounts_de.db
├── persisted_taskIds.txt
├── ringtones
└── spblob
```

DB Browser for SQLite - /home/kali/pixel3/Pixel 3/data/system_de/0/accounts_de.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Open Project Save Project

Database Structure Browse Data Edit Pragma Execute SQL

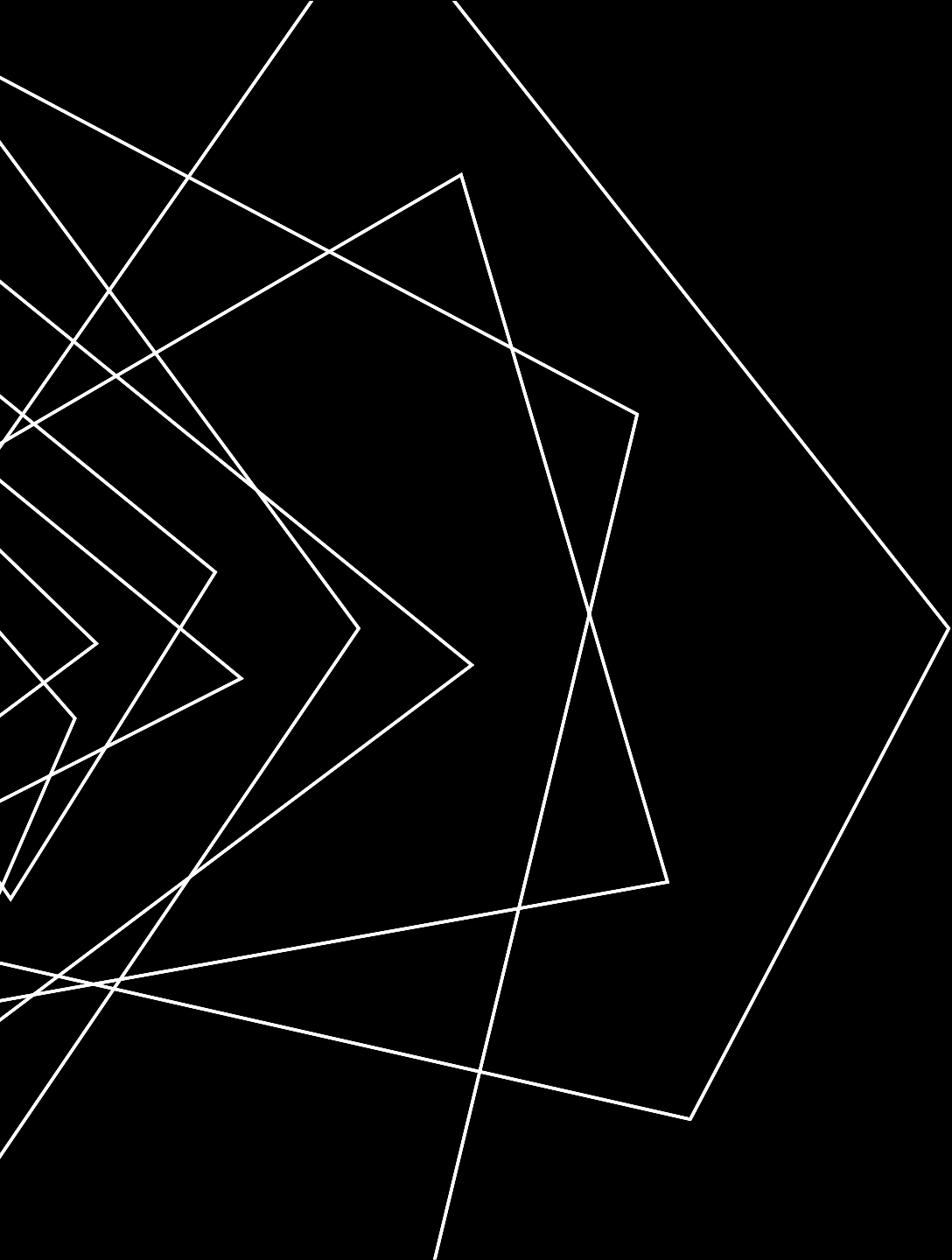
Table: accounts

	_id	name	type	previous_name	last_password_entry_time_millis_ep
1	1	thisisdfr@gmail.com	com.google	NULL	1580312467984
2	2	858233690	org.telegram.messenger	NULL	1580322664981
3	3	Signal	org.thoughtcrime.securesms	NULL	1580322700050
4	4	imo HD	com.imo.android.imoos	NULL	1580322720395
5	5	TikTok	com.zhiliaoapp.musically	NULL	1580323396517
6	6	Messenger	com.facebook.messenger	NULL	1580323862026
7	7	TDfir	com.twitter.android.auth.login	NULL	1580326971888
8	8	WhatsApp	com.whatsapp	NULL	1580332202948
9	10	thisisdfr	com.silentcircle.account	NULL	1580589691083
10	11	Skype	com.skype.raider	NULL	1580591300780
11	12	TextNow	com.enflick.android.TextNow.account	NULL	1581191692330
12	13	+19195794674	com.viber.voip	NULL	1581192938951
13	14	Duo	com.google.android.apps.tachyon	NULL	1581643155834

need convert *last password entry time* to human readable

Convert *last password entry time millis epoch to human* readable

```
(kali㉿kali)-[~/pixel3]
$ sqlite3 'Pixel 3/data/system_de/0/accounts_de.db' 148 x 1 ⚙
SQLite version 3.33.0 2020-08-14 13:23:32
Enter ".help" for usage hints.
sqlite> select name, type, datetime(last_password_entry_time_millis_epoch/1000, 'UNIXEPOCH') from accounts;
thisisdfir@gmail.com|com.google|2020-01-29 15:41:07 ←
858233690|org.telegram.messenger|2020-01-29 18:31:04
Signal|org.thoughtcrime.securesms|2020-01-29 18:31:40
imo HD|com.imo.android.ious|2020-01-29 18:32:00
TikTok|com.zhiliaoapp.musically|2020-01-29 18:43:16
Messenger|com.facebook.messenger|2020-01-29 18:51:02
TDfir|com.twitter.android.auth.login|2020-01-29 19:42:51
WhatsApp|com.whatsapp|2020-01-29 21:10:02
thisisdfir|com.silentcircle.account|2020-02-01 20:41:31
Skype|com.skype.raider|2020-02-01 21:08:20
TextNow|com.enflick.android.TextNow.account|2020-02-08 19:54:52
+19195794674|com.viber.voip|2020-02-08 20:15:38
Duo|com.google.android.apps.tachyon|2020-02-14 01:19:15 ←
sqlite> █
```



THANK YOU

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